

**Tutorial 4 Discussion**

1. The four different methods that can be used to perform a deployment are: Manual, Scripted, Language based, and Model based. Which method will you use for the following scenarios:
  - a. Your final year team project -- for the test environment, for final submission.
  - b. A web browser like Firefox that is deployed/used in millions to machines.
  - c. A simple PHP based web application -- it has a set of scripts, some Apache configuration, MySQL server.
2. What are the parameters that you used in order to determine the deployment method?
3. Consider a PHP based e-commerce web application again. It has 200 PHP pages, 300 static HTML pages, 20 CSS files, 50 Javascript files, 100 images (average size 500Kb), 400Mb database (MySQL), running on the latest version of Apache+PHP. The entire application is hosting on a single physical machine in US, the development team is based on Melbourne, AU.
  - a. How will you deploy this application? (Assume different types of files will go into different directories -- and you have full shell access to the server).
  - b. What is your packaging format that would be used?
4. Assume the same scenario as in Question 3.
  - a. What are the types of questions that you would ask in order to ensure that you can adjust your deployment process? (e.g. Scalability, Frequency of deployment, Training of team...)
5. Assume the same scenario as in Question 3. This e-commerce application is sold in a SaaS model (Software as a Service) to a number of different clients. That is, many small businesses use the same application to host their online commerce sites. Unfortunately, each time the sales team makes a sale -- the initial setup requires modification to the PHP code base, an update to the database and creation of 20-30 customer-specific images/HTML/CSS. The sales team on average sells to 10 clients every month.
  - a. Will this new information influence any choices you make in the deployment? (You cannot re-write the PHP application, as there is no risk appetite or budget available for such work)
6. Assume scenario from question 5.
  - a. Will you use a language to perform deployments or will a script suffice?
7. Assume scenario from Questions 3/5.
  - a. How many different environments will you run in the development?
  - b. That is, will you recommend Dev/Test/Staging and Live environments (or) would this be overkill?
8. Assume the PHP web application scenario. Due to the increasing popularity of web shopping during a holiday season you now have the need to provide additional

physical servers during holiday season (for 6 weeks). During this period, in order to minimize costs, the business will lease an additional 2 servers (taking the total to 3 physical machines). 1 server runs the database, while 2 servers run the front-end Apache web server. The hosting provider also gives you a load balancer to manage traffic -- this load balancer needs some additional configuration.

- a. What are the parameters that you will use to make a decision on the level of automation for the deployment?